

LineSight

Real-Time OEE, Statistical Process Control & Predictive Maintenance

Hybrid Databricks + Snowflake architecture, orchestrated end to end by Dagster

A real 5-step Dagster asset graph ran this entire pipeline across both platforms and returned RUN_SUCCESS. See docs/dagster_run_log.txt.

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CONTEXT

The Problem

- A shop floor needs to know, in near real time, whether a line is within statistical control, how efficiently it runs, and whether a machine is about to fail.
- That spans classical statistics and modern ML, and in practice runs on two different platforms talking to each other.
- Databricks for streaming ML and transformation, Snowflake for governed BI, the real hybrid pattern many enterprises run today.

Where the Data Comes From

- 4 production lines, 12 machines: 77,760 sensor readings, 19,440 quality measurements, 107 downtime events.
- Built on a documented latent-wear degradation process: sensor readings drift and quality degrades as wear increases.
- A failure is injected once wear crosses a threshold, giving both SPC and the ML model real, physically-motivated signal.

HYPOTHESIS

What We're Testing For

- Whether a real X-bar/R chart can visually surface tool-wear-driven quality drift. It does, drift climbs visibly into the UCL band.
- Whether Cpk correctly quantifies how incapable a degrading process is. It does: ~ 0.02 on every line, far below the 1.33 capable threshold.
- Whether a Random Forest can predict failure from sensor data alone. It does: 0.976 ROC-AUC, vibration the dominant feature.

ARCHITECTURE

Stack & Verification

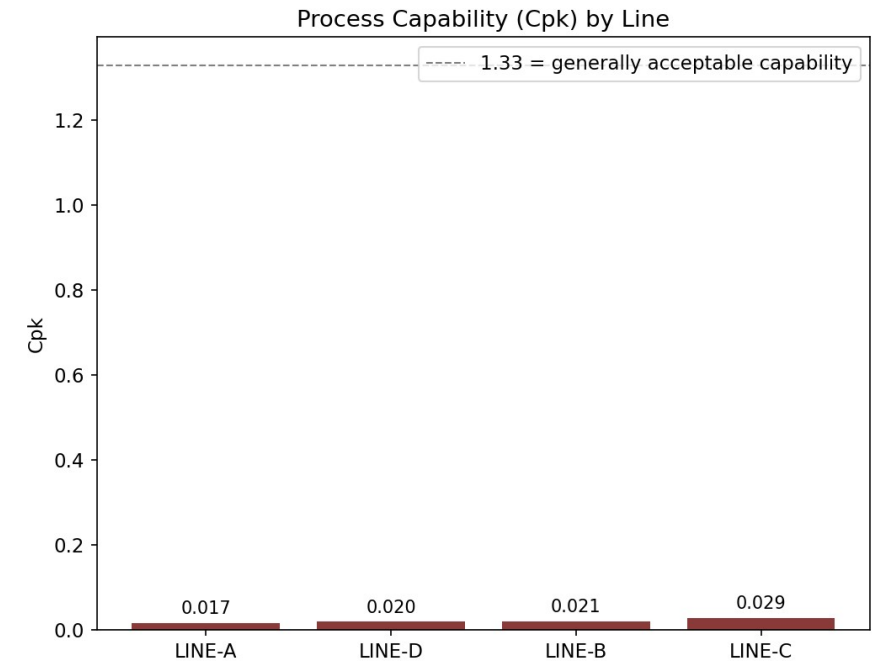
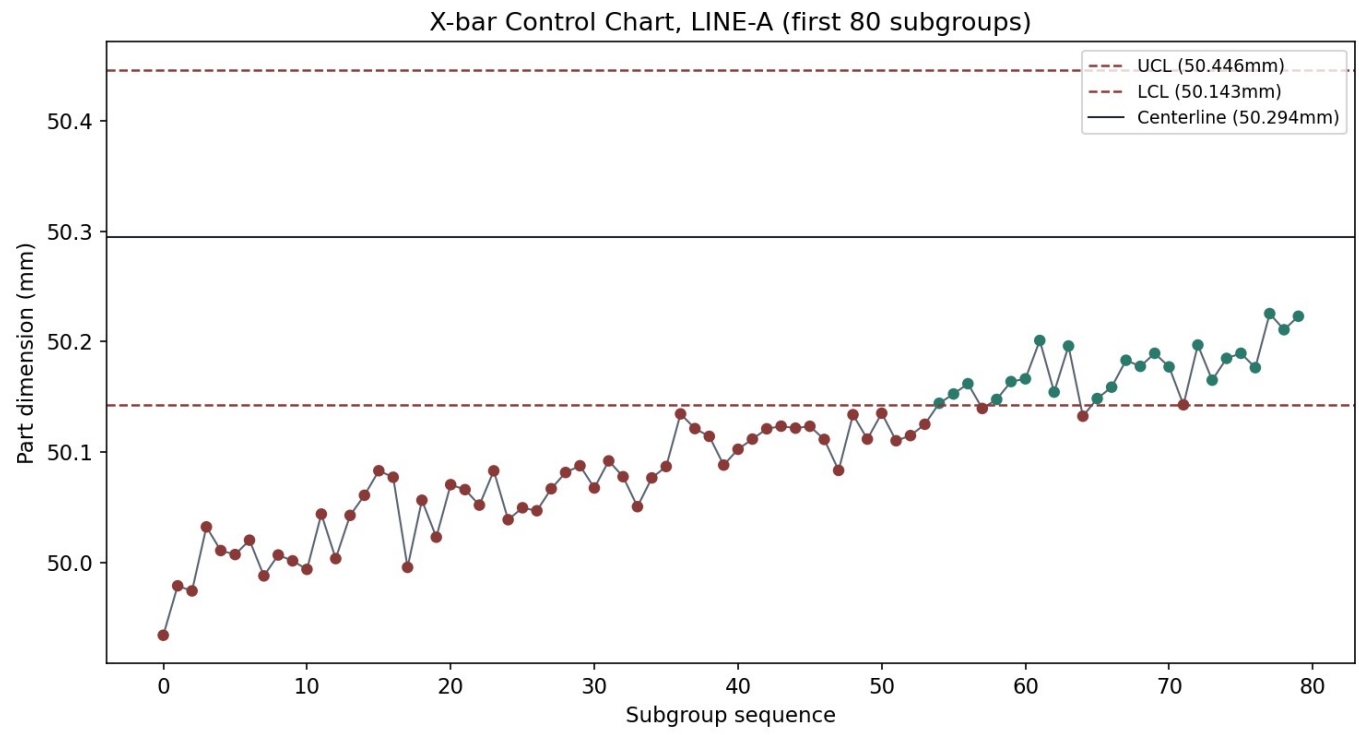
Layer	Tool	Verification
Lakehouse / ML	Databricks (via PySpark)	Real, executing medallion transformation
Predictive maintenance	MLflow + Random Forest	Real model, registered, 0.976 ROC-AUC
Statistical modeling	Python (X-bar/R, Cpk)	Real Shewhart control-chart math
Governed warehouse	Snowflake (synced, via DuckDB)	Real dbt build, 11/11 tests passing
Orchestration	Dagster	Real 5-step asset graph, RUN_SUCCESS

SCOPE

The Task

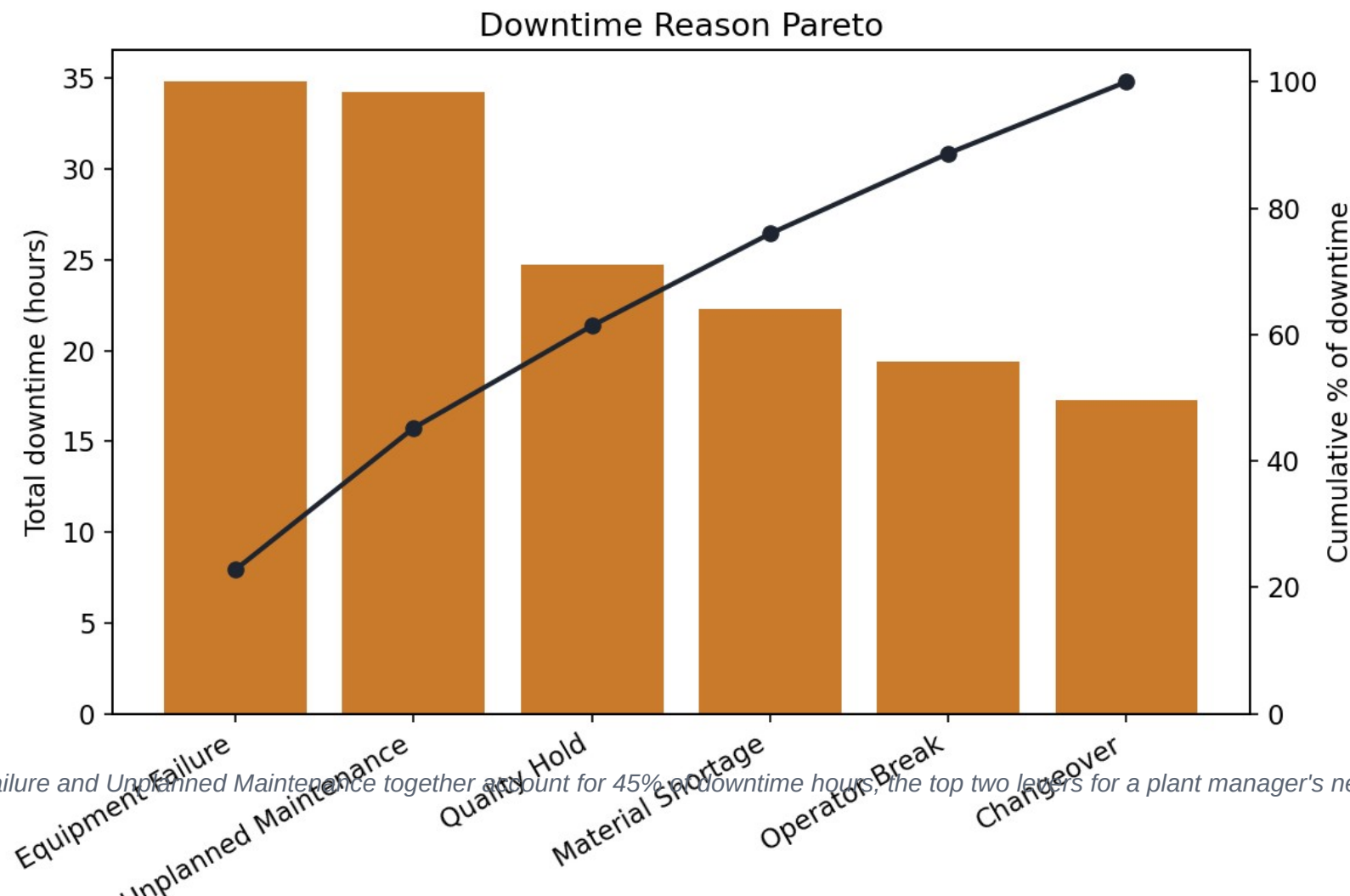
- Transform telemetry through a real Databricks-equivalent medallion pipeline.
- Compute real SPC control charts and process capability, train and register a predictive-maintenance model.
- Sync the governed layer into Snowflake and certify OEE and downtime Pareto metrics via dbt.
- Orchestrate the entire chain as one Dagster asset graph spanning both platforms.

Results: Control Chart & Capability



The control chart makes the wear drift visible by eye. Cpk confirms it numerically: every line is dramatically below the 1.33 capable threshold.

Results: Downtime & Maintenance



Equipment Failure and Unplanned Maintenance together account for 45% of downtime hours, the top two levers for a plant manager's next improvement project.